

Diagnostic report: Mg I

Computed levels: 25

Matched to NIST: 15

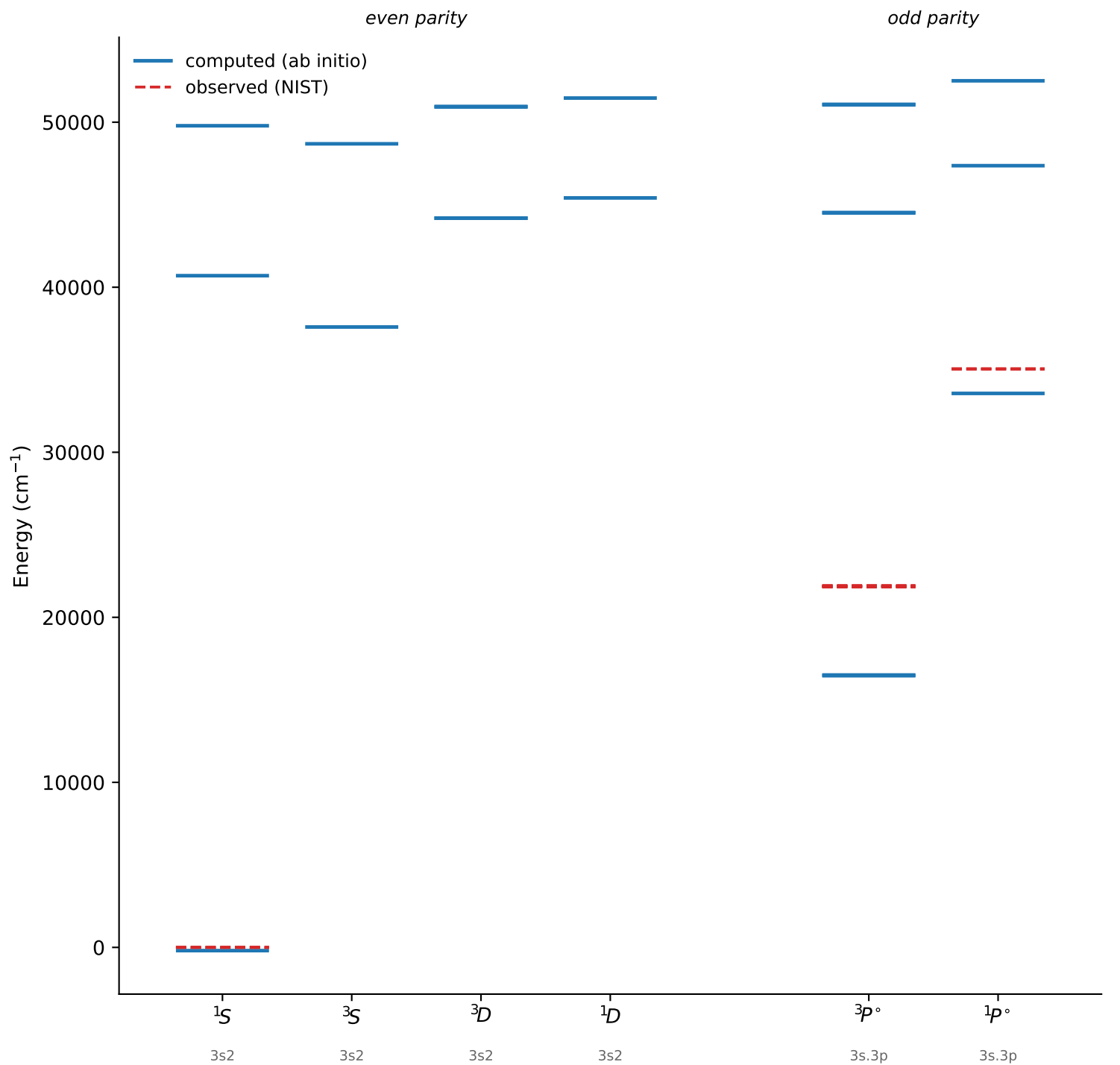
Computed E1 lines: 52

Pre-fit RMS ($E_{\text{calc}} - E_{\text{obs}}$): 24186 cm^{-1} (ab initio, no RCE fit yet)

Strongest computed lines:

lambda(A)	log gf	upper term
24404.97	+0.595	(2S) 1P
42031.78	+0.507	(2S) 3P
3612.66	+0.470	(2S) 3P
15581.33	+0.424	(2S) 3P
8441.88	+0.421	(2S) 1P
2962.04	+0.393	(2S) 1P
14430.30	+0.382	(2S) 3P
42069.26	+0.285	(2S) 3P

Mg I: term (Grotrian) diagram



Mg I: levels (computed vs observed)

config	term	J	E_calc	E_obs	ΔE (cm ⁻¹)
3s2	¹ S	0	-195.0	0.0	-195.0
3s2	¹ S	0	40700.0	0.0	+40700.0
3s2	¹ S	0	49785.0	0.0	+49785.0
3s.3p	³ P°	0	16463.0	21850.4	-5387.4
3s.3p	³ P°	1	16477.0	21870.5	-5393.5
3s.3p	³ P°	2	16505.0	21911.2	-5406.2
3s.3p	³ P°	0	44511.0	21850.4	+22660.6
3s.3p	³ P°	1	44514.0	21870.5	+22643.5
3s.3p	³ P°	2	44519.0	21911.2	+22607.8
3s.3p	³ P°	0	51066.0	21850.4	+29215.6
3s.3p	³ P°	1	51067.0	21870.5	+29196.5
3s.3p	³ P°	2	51070.0	21911.2	+29158.8
3s.3p	¹ P°	1	33566.0	35051.3	-1485.3
3s.3p	¹ P°	1	47365.0	35051.3	+12313.7
3s.3p	¹ P°	1	52508.0	35051.3	+17456.7
3s2	³ S	1	37590.0	--	--
3s2	³ S	1	48690.0	--	--
3s2	³ D	1	44185.0	--	--
3s2	³ D	2	44185.0	--	--
3s2	³ D	3	44185.0	--	--
3s2	³ D	1	50937.0	--	--
3s2	³ D	2	50937.0	--	--
3s2	³ D	3	50937.0	--	--
3s2	¹ D	2	45411.0	--	--
3s2	¹ D	2	51462.0	--	--

Mg I: level-fit residuals (RMS = 24186 cm⁻¹, ab initio)

